

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 10: Satellite Integrity Monitoring (RAIM & SBAS/GBAS)

Syllabus Source: CAE Oxford (pp. 385-424) | Pass Target: >= 90% AviationExam

1. RECEIVER AUTONOMOUS INTEGRITY MONITORING (RAIM)

GPS satellites can transmit erroneous ranging signals without immediate warning. RAIM is an internal airborne algorithm that cross-checks satellite ranges for consistency:

- Fault Detection (FD): REQUIRES A MINIMUM OF 5 SATELLITES (or 4 satellites + Barometric VNAV aiding). Detects if a faulty satellite is providing corrupt data, alerting the pilot with an integrity warning.
- Fault Detection and Exclusion (FDE): REQUIRES A MINIMUM OF 6 SATELLITES (or 5 satellites + Baro-VNAV). Identifies WHICH satellite is faulty, excludes it from the navigation solution, and continues navigation without interruption!
- Mask Angle: Satellites within 5° of the horizon are rejected (masked) due to excessive tropospheric refraction.

2. SATELLITE-BASED AUGMENTATION SYSTEMS (SBAS)

SBAS uses ground reference networks and geostationary satellites to broadcast wide-area differential corrections:

- Systems: EGNOS (Europe), WAAS (USA), MSAS (Japan), GAGAN (India).
- Capability: Eliminates ionospheric delay errors. Enables LPV (Localizer Performance with Vertical Guidance) approaches down to ILS Cat I decision height (200 ft) without ground ILS equipment!

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the MINIMUM number of operational satellites required for RAIM to perform FAULT DETECTION (FD)?

[A] 4 satellites [B] AT LEAST 5 SATELLITES (or 4 with Barometric Altimeter aiding) [C] 6 satellites [D] 3 satellites

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Fault Detection requires 5 visible satellites with suitable geometry to detect an anomalous pseudo-range error.

Q2: How many satellites are required for RAIM to perform FAULT DETECTION AND EXCLUSION (FDE)?

[A] 5 satellites [B] AT LEAST 6 SATELLITES (or 5 with Baro-VNAV aiding) [C] 8 satellites [D] 4 satellites

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. FDE requires 6 satellites: 4 to solve the 3D position/time, a 5th to detect the fault, and a 6th to isolate and exclude the corrupt satellite.

Q3: What is the European Satellite-Based Augmentation System (SBAS) called?

[A] WAAS [B] EGNOS (European Geostationary Navigation Overlay Service) [C] Galileo [D] GLONASS

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. EGNOS is the European SBAS system that augments GPS signals to provide safety-of-life LPV approach capabilities.